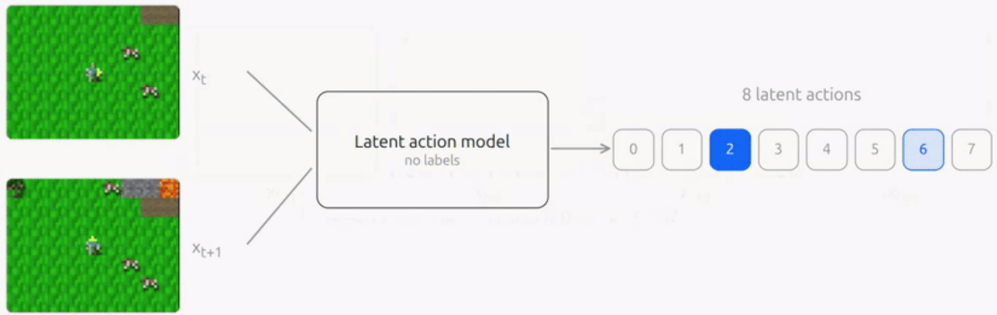
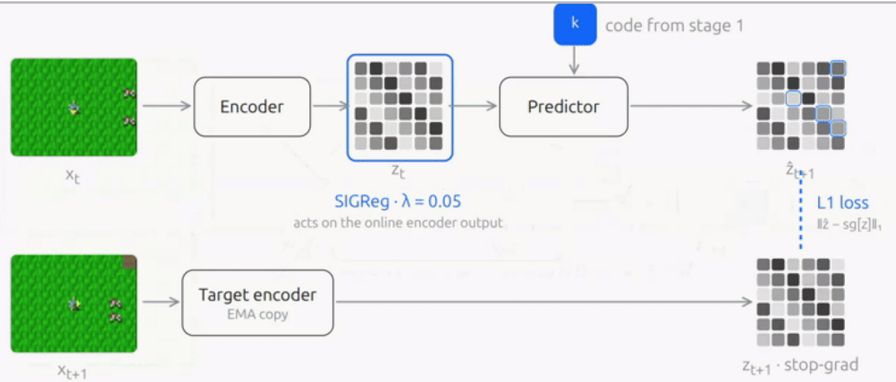
(a) Stage 1: Unsupervised Action Discovery (LAM, 0% Labels)



Each change between frames becomes one of 8 discrete codes.

Illustrative · the codes have no names yet.

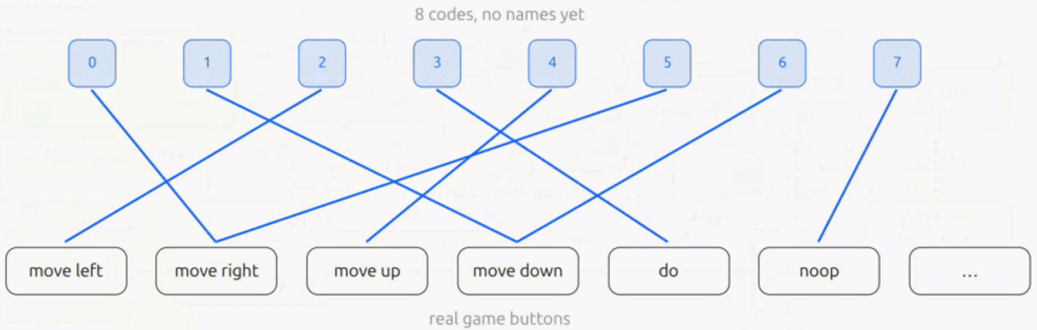
(b) Stage 2: Latent State Prediction & SIGReg (0% Labels)



It predicts the next frame's representation, not its pixels.

Loss = L1($\hat{z}$, sg[z]) + 0.05 · SIGReg(z_t) + 0.02 · variance hinge

(c) Stage 3: Few-Shot Action Alignment Probe (1% Labels)

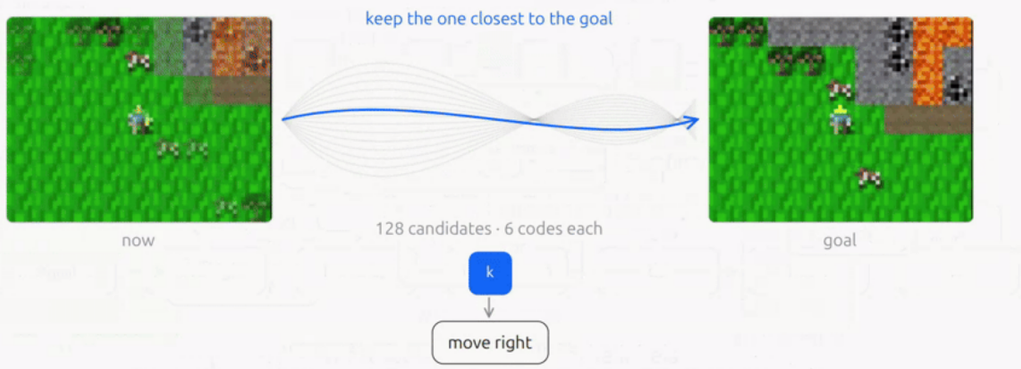


44.7% accuracy with 1% of the labels

A few labelled frames are enough to name the codes.

A tiny probe is trained; everything else stays frozen.

(d) Stage 4: Closed-Loop Latent CEM Planning (All Frozen)



Take the first step, then plan again.